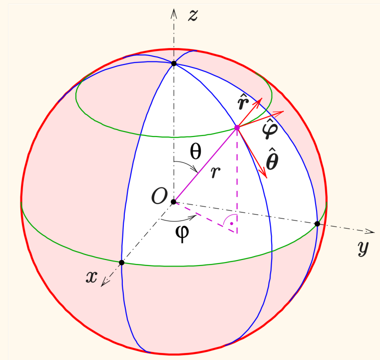
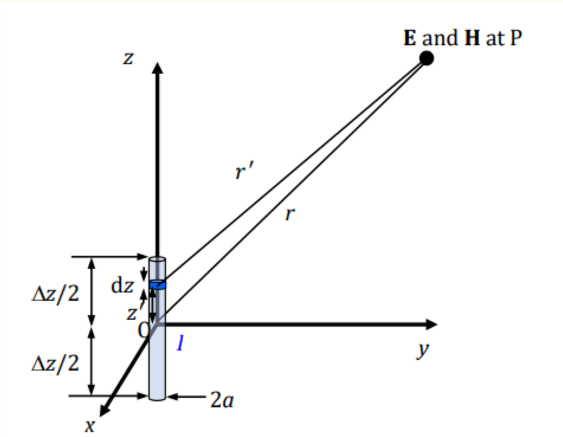
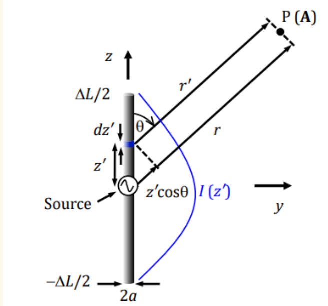
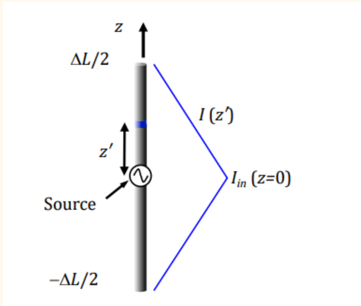


$$\eta = \frac{|E|}{|H|} = \frac{\omega\mu}{k} \qquad R_{rad} = \frac{2P_{rad}}{|I|^2} \qquad R_{rad} + R_{loss} = \frac{2P_{in}}{|I|^2}$$

<table>
<tr><th>Hertzian Dipole</th><th>Short Dipole (L < 0.1λ)</th></tr>
<tr><td>


(a) Hertzian dipole in a spherical coordinate system
</td><td>


A center-fed short dipole antenna



A short dipole with triangular current

For a short dipole $\Delta L<\lambda/10$,

$$\mathbf{I}(z') = \begin{cases} \mathbf{a}_z I_{in}\left(1 - z'\frac{2}{\Delta L}\right), & 0 \le z' \le \frac{\Delta L}{2} \\ \mathbf{a}_z I_{in}\left(1 + z'\frac{2}{\Delta L}\right), & -\frac{\Delta L}{2} \le z' \le 0 \end{cases}$$
</td></tr>
<tr><td>
$$E = a_\theta E_\theta = a_\theta j\omega\mu L I \frac{e^{-jkr}}{4\pi r}\sin\theta$$
$$H = a_\phi H_\phi = a_\phi jkLI \frac{e^{-jkr}}{4\pi r}\sin\theta$$
$$V = j\omega\mu L I \frac{1}{4\pi}$$
</td><td>
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$$F_{\theta, Hertzian\ Dipole}(\theta, \phi) = \sin(\theta)$$
$$F_{\phi, Hertzian\ Dipole}(\theta, \phi) = 0$$
</td><td>
$$F_\theta(\theta, \phi) = \sin(\theta)$$
$$F_\phi(\theta, \phi) = 0$$
</td></tr>
<tr><td>
$$S_{ave} = \frac{1}{8}\eta\left(I\frac{L}{\lambda}\right)^2 \frac{\sin^2\theta}{r^2} a_r$$
$$P_{rad} = \frac{\pi}{3}\eta\left(I\frac{L}{\lambda}\right)^2$$
$$R_{rad} = \frac{2\pi}{3}\eta\left(\frac{L}{\lambda}\right)^2$$
HPBW = 90º
</td><td>
$$S_{ave} = \frac{1}{32}\eta\left(I\frac{L}{\lambda}\right)^2 \frac{\sin^2\theta}{r^2} a_r$$
$$P_{rad} = \frac{\pi}{12}\eta\left(I\frac{L}{\lambda}\right)^2$$
$$R_{rad} = \frac{\pi}{6}\eta\left(\frac{L}{\lambda}\right)^2$$
$$X_{in} = \frac{-60\lambda}{\pi L}\left(\ln\left(\frac{L}{a}\right) - 1\right)$$
HPBW = 90º
</td></tr>
</table>

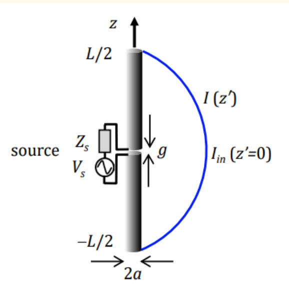
$U(\theta, \phi) = \frac{1}{8}\eta(I\frac{L}{\lambda})^2 sin^2\theta$	$U(\theta, \phi) = \frac{1}{32}\eta(I\frac{L}{\lambda})^2 sin^2\theta$
$U_{max} = \frac{1}{8}\eta(I\frac{L}{\lambda})^2 = \frac{\lvert V\rvert^2}{2\eta}$	$U_{max} = \frac{1}{32}\eta(I\frac{L}{\lambda})^2 = \frac{\lvert V\rvert^2}{2\eta}$
$V = j\omega\mu LI\frac{1}{4\pi}$	$V = j\omega\mu LI\frac{1}{8\pi}$
$D(\theta, \phi) = \frac{3sin^2\theta}{2}$	$D(\theta, \phi) = \frac{3sin^2\theta}{2}$
$D = 1.5 = 1.76\, dBi$	$D = 1.5 = 1.76\, dBi$

The isotropic gain, G_i, of a half-wave dipole is 1.64, or in decibels 10 log 1.64 = 2.15 dBi

- if datasheet of halfwave dipole gives equal or higher than that means sussy

Summary

Dipole



$$\mathbf{I}(z') = \begin{cases} \mathbf{a}_z I_{in} \left|\cos\left[k\left(z' - \frac{L}{2} + \frac{\lambda}{4}\right)\right]\right| & \text{for } 0 \le z' \le \frac{L}{2} \\ \mathbf{a}_z I_{in} \left|\cos\left[k\left(z' + \frac{L}{2} + \frac{\lambda}{4}\right)\right]\right| & \text{for } -\frac{L}{2} \le z' \le 0 \end{cases}$$

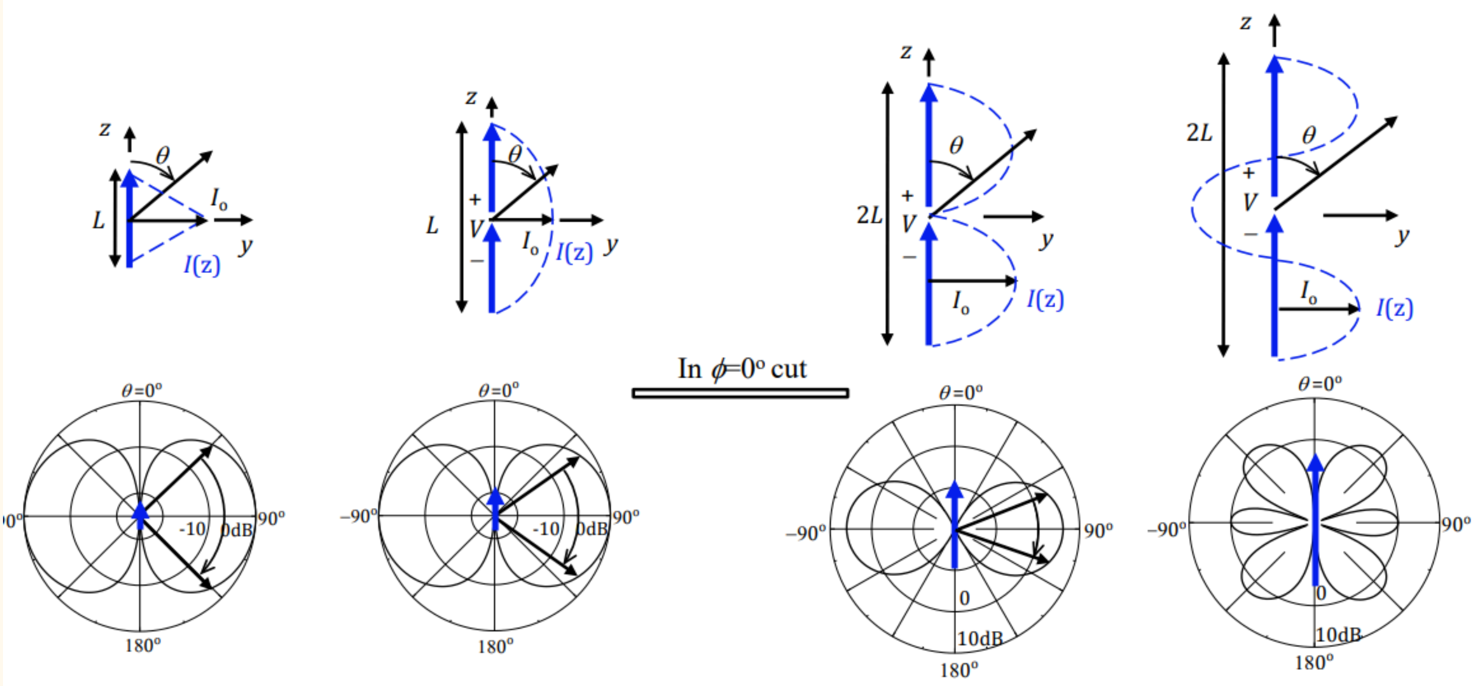
$$\begin{cases} dE_\theta = j60\pi \frac{I(z')\sin\theta}{\lambda r'} dz' & \text{V/m} \\ dH_\phi = j\frac{I(z')\sin\theta}{2\lambda r'} dz' & \text{A/m} \end{cases}$$

$$E = a_\theta E_\theta = a_\theta jI\eta \frac{e^{-jkr}}{2\pi r} [\frac{cos(kL/2\, cos\theta) - cos(kL/2)}{sin\theta}]$$

$$H = a_\phi H_\phi = a_\phi jI \frac{e^{-jkr}}{2\pi r} [\frac{cos(kL/2\, cos\theta) - cos(kL/2)}{sin\theta}]$$

$$F_\theta(\theta, \phi) = \frac{cos(kL/2\, cos\theta) - cos(kL/2)}{sin\theta}$$

$$F_\phi(\theta, \phi) = 0$$

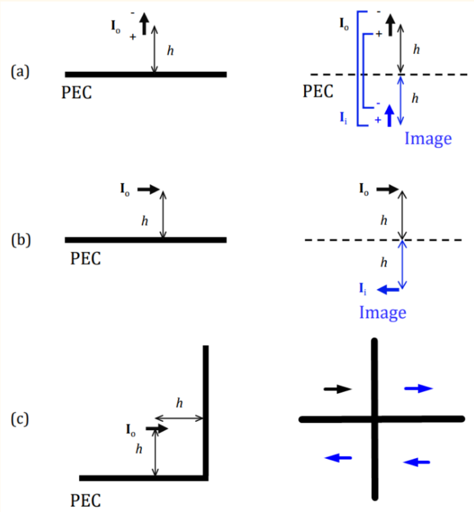


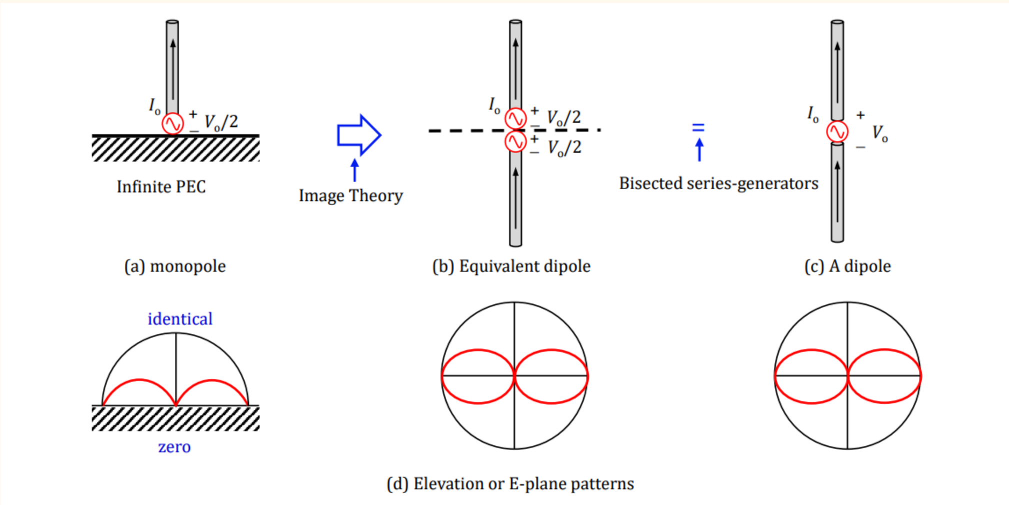
Length L	$<< \lambda$	$\lambda/2$	λ	$3\lambda/2$
Radiation pattern $F_\theta(\theta)$	$\sin\theta$	$\dfrac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}$	$\dfrac{\cos(\pi\cos\theta)+1}{\sin\theta}$	$\dfrac{\cos\left(\frac{3}{2}\pi\cos\theta\right)}{\sin\theta}$
Input resistance R_{in} (Ω)	$80\pi^2(L/\lambda)^2$	73.1	-	-
HPBW	90°	78°	47°	
Directivity D	1.5 or 1.76 dBi	1.64 or 2.15 dBi	3.28 or 5.15 dBi	1.64 or 2.15 dBi at θ=90°

5 Chen Zhi Ning, NUS

- larger antenna up to full wave gives the strongest directivity
- As length increases, resonance frequency decreases
  - dipole resonate at half wavelength (optimised is 0.42λ)
  - antenna behaves more inductive at fixed frequency
- for the fixed length, directivity increases as frequency increase, as antenna is larger electrically (as long it's still before 3λ/2), but realised gain decrease due to mismatch efficiency

<u>Monopole</u>



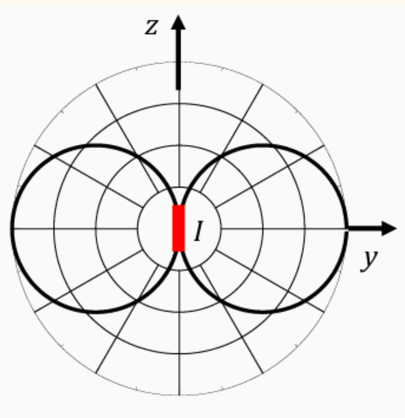


This is for the monopole with length = half the dipole

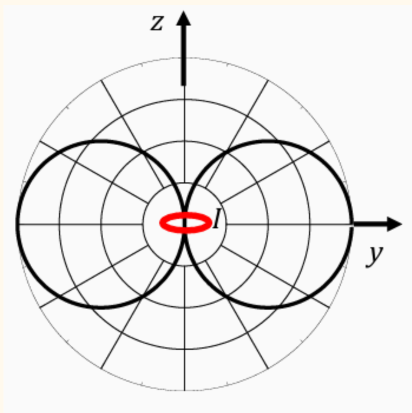
$$D_{monopole}(\theta,\ \phi) = \frac{U_{max,\,monopole}}{U_{ave,\,monopole}} = \frac{U_{max,\,dipole}}{\frac{1}{2}U_{ave,\,dipole}} = 2D_{dipole}(\theta,\ \phi) \quad \{0 < \theta < 90\}$$

$$Z_{monopole} = \frac{V_{in,\,monopole}}{I_{in,\,monopole}} = \frac{\frac{1}{2}V_{in,\,dipole}}{I_{in,\,dipole}} = \frac{1}{2}Z_{dipole}$$

Dipole Loop Duality



(a) Linearly θ-polarized



(b) Linearly ϕ-polarized

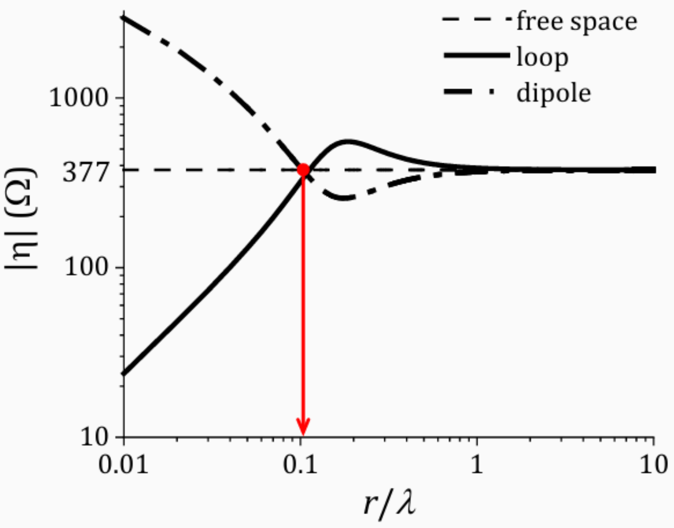
Dipole: $\begin{cases} H_\phi = jkI_e l \dfrac{e^{-jkr}}{4\pi r}\sin\theta \ \text{A/m} \\ E_\theta = j\eta k I_e l \dfrac{e^{-jkr}}{4\pi r}\sin\theta \ \text{V/m} \end{cases}$

Dipole: $\begin{cases} \mathbf{H} = \mathbf{a}_\phi H_\phi \ \text{A/m} \\ \mathbf{E} = \mathbf{a}_\theta E_\theta \ \text{V/m} \end{cases}$

Loop: $\begin{cases} H_\theta = -k^2 I_e A \dfrac{e^{-jkr}}{4\pi r}\sin\theta \ \text{A/m} \\ E_\phi = \eta k^2 I_e A \dfrac{e^{-jkr}}{4\pi r}\sin\theta \ \text{V/m} \end{cases}$

Loop: $\begin{cases} \mathbf{H} = \mathbf{a}_\theta H_\theta \ \text{A/m} \\ \mathbf{E} = \mathbf{a}_\phi E_\phi \ \text{V/m} \end{cases}$

Using duality, transform electric (E, H, I_e, ε, μ, η) and magnetic (H, $-E$, I_m, μ, ε, $1/\eta$) quantities one to one. Small loop acts as an infinitesimal magnetic dipole: $I_m l = jA\omega\mu I_e$.



(a) Wave impedance vs distance

- *Wave Impedance* (lossless): $|\eta| = \left|\frac{\mathbf{E}}{\mathbf{H}}\right|\ \Omega$ and $a = \frac{1}{kr}$.
  - Hertzian dipole: $|\eta| = \left|\frac{E_\theta}{H_\phi}\right| = 120\pi \left|\frac{\sqrt{1+a^6}}{1-a^2+a^4}\right|\ \Omega$
  - Small loop: $|\eta| = \left|\frac{E_\phi}{H_\theta}\right| = 120\pi \left|\frac{1-a^2+a^4}{\sqrt{1+a^6}}\right|\ \Omega$
- A special distance from antenna: $r = \frac{\lambda}{2\pi}$.
- In nearfield zone($r \ll 0.1\lambda$), majority of energy generated is
  - Hertzian dipole: electric energy with larger $|\eta|$ or $|E|$.
  - Small coil: magnetic energy with smaller $|\eta|$ or larger $|H|$.